



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-2027

Course Handout

Date: 1/8/2026

In addition to the part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : PHY F313
Course Title : Computational Physics
Instructor-in-Charge : Tanmoy Mondal
Instructors : Tanmoy Mondal

1. Course Description:

The course introduces elementary computational methods for obtaining approximate solutions to mathematical problems arising in physics. Topics include root finding, systems of linear equations, numerical quadrature, ordinary and partial differential equations, Monte Carlo simulation, and Markov Chain Monte Carlo methods. Emphasis will be placed on the hands-on implementation of basic algorithms and their application to representative physical systems.

2. Scope and Objective of the Course:

The objective of the course is to develop a practical understanding of the basic numerical methods used to obtain approximate solutions of mathematical problems arising in physics. Emphasis will be placed on the hands-on construction of elementary algorithms, the translation of equations into finite computational procedures, and the interpretation and validation of numerical results. Students will also learn to use scientific programming and visualization tools, including Monte Carlo and Markov Chain Monte Carlo techniques, for studying relevant physical systems.

3. Textbooks:

1. *Computational Physics*, N.J. Giordano, H. Nakanishi, Pearson Prentice Hall, II edition, 2006
2. *An Introduction to Computer Simulation Methods*, Harvey Gould, Jan Tobochnik & Wolfgang Christian, Addison Wesley, 3rd Edition, 2005

4. Reference Books:

1. *An Introduction to Computational Physics*, Tao Pang, Cambridge University Press, II edition, 2008



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2. *An Introduction to Computational Physics*, Paul DeVries, John Wiley & Sons, I Indian edition, 2011
3. *MCMC from Scratch: A practical introduction to Markov Chain Monte Carlo*, M. Hanada and S. Matsuura, Springer
4. *Numerical Methods for Engineers*, Steven C. Chapra & Raymond P. Canale, McGraw Hill; 8th edition (2021)

5. Course Plan

Week No	Module No.	Lecture/Tutorial Session.	Reference	Learning Outcome
1	1. Root finding (Lectures: 1-2)	Bisection Newton-Raphson	Class Notes	Implement and apply bracketing and NR methods to equations arising in physics.
2-3	2. Numerical integration (Lectures: 3-8)	Simpson's Rule Gaussian Quadrature	TB1: Ch 1 TB2: Ch 11 Class Notes	Construct, compare and examine the convergence of quadrature rules.
3-4	3. Systems of linear equations (Lectures: 9-12)	Gaussian Elimination Jacobi and Gauss-Seidel Iterative methods	Class Notes	Learn solving systems of linear equations using direct & iterative methods
4-7	4. Ordinary differential equations (Lectures: 13-21)	1 st Order ODEs Euler, Runge-Kutta Methods 2 nd Order ODEs Shooting Method	TB1: Ch 6 Class Notes	Convert physical equations of motion into first-order systems and obtain numerical trajectories.
8-9	5. Partial Differential equations (Lectures: 22-25)	Laplace equation Finite-difference method	Class Notes	Discretize Laplace's equation and obtain numerical solutions using finite difference method.
9-11	6. Monte Carlo methods (Lectures: 26-33)	Random-number generation, Random sampling, Monte Carlo integration, Importance sampling.	TB1: Ch 7 R1: Ch 5 R3: Ch 2	Construct Monte Carlo estimators and examine their statistical convergence.



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Week No	Module No.	Lecture/Tutorial Session.	Reference	Learning Outcome
12-13	7. Markov Chain Monte Carlo (Lectures: 34-38)	General aspects of MCMC, Metropolis algorithm, Metropolis–Hastings algorithm	R3: Chs 3 - 5	Explain and implement elementary Metropolis and Metropolis–Hastings sampling.
13-14	8. Applications (Lectures: 39-42)	Ising Model, Bayesian inference	R3: Ch 6	Apply MC to selected physical models.

6. Evaluation Scheme:

Component	Duration	Marks (Weightage %)	Date & Time	Nature of component (Closed Book/ Open Book)
Mid-Semester Examination	90 min	90 (30%)	TBA	Closed
Comprehensive Examination	180 min	120 (40%)	TBA	Closed + Open
Class Tests (best 3 out of 4)	30 min x 4	30 marks each; best 3 = 90 (30%)	TBA	Closed

7. Consultation Hour: To be announced in the class.

8. Notices: These will be displayed only on the Google Classroom/Nalanda website.

9. Make-up Policy: No make-up for class tests. Make-up for other components will be granted only to genuine cases with prior intimation and permission as per Institute rules.

10. It is the responsibility of the students to keep all the evaluated and returned answer sheets till the end of the semester.

Tannoy Mondal .

Instructor-in-charge
Course No. PHY F313



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